

Literature status: the polyhedral subcase of property B

Run `fa_banach_001`; source arXiv:1306.1155

Source question

Miguel Martin's paper *Norm-attaining compact operators* asks the following finite-rank density question.

$$\|P_2 S_0 - P_2 S\| \leq \|S_0 - S\| < \varepsilon,$$

as claimed.

Now, if we take X , Y , and $T_0 \in K(X, Y)$ which cannot be approximated by norm-attaining operators, then $Z = X \oplus_\infty Y$ and $S_0 \in K(Z, Z)$ defined as above, give the desired example. $\square$

Let us finish the paper with an small discussion about positive results on norm-attaining compact operators. The main open question here (and also in the general theory of norm-attaining operators) is whether finite-rank operators can be always approximated by norm-attaining operators.

Question 9. Let Y be a finite-dimensional space. Is it true that for every Banach space X , $NA(X, Y)$ is dense in $L(X, Y)$?

Figure 1: Question 9 in arXiv:1306.1155, page 4.

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We would like to comment that the problem above is open even when Y is the 2-dimensional real Hilbert space. Related to this, let us also comment that there is a complex version of the Bishop-Phelps theorem which states that for every complex Banach space X , complex-linear norm-attaining operators (i.e. functionals) from X into $\mathbb{C}$ are dense in the space of all complex-linear operators (see [19] or [20, §2]). On the other hand, V. Lomonosov showed in 2000 [18] that there is a complex Banach space X and a (non-complex symmetric) closed convex bounded subset C of X such that there is no element in X^* attaining the supremum of its modulus on C .

Figure 2: The following page notes that the problem is open even for the two-dimensional real Hilbert space.

Question 9 asks whether, for every finite-dimensional range space Y and every Banach space X , the set $NA(X, Y)$ is dense in $\mathcal{L}(X, Y)$.

Supporting literature status

The polyhedral finite-dimensional range subcase is already known. Kadets, Lopez, Martin, and Werner's 2019 paper *Norm attaining operators of finite rank* recalls Lindenstrauss's examples of spaces with property B , including finite-dimensional polyhedral spaces.

The same paper also says that the all finite-dimensional version remains a major open question.

Thus the statement below is a literature-known proper subcase of Martin's Question 9, not a new partial result from this run and not a solution of the full question.

In particular, we consider the case of Hilbert space valued operators where we obtain a complete characterization of these properties. In the final section we offer a candidate for a counterexample to the complex Bishop-Phelps theorem on c_0 , the first such counterexample on a certain complex Banach space being due to V. Lomonosov.

1. INTRODUCTION

Shortly after Bishop and Phelps's papers ([6], [7]) on the density of norm attaining functionals on a Banach space had appeared, Lindenstrauss, in his seminal work [24], launched the study of norm attaining operators. Let us recall that a bounded linear operator $T: X \rightarrow Y$ between Banach spaces, $T \in \mathcal{L}(X, Y)$, is called *norm attaining* if there is some $x_0 \in X$ with $\|x_0\| = 1$ and $\|Tx_0\| = \|T\|$; in this case we write $T \in \text{NA}(X, Y)$. Lindenstrauss introduced the following properties (A) and (B) of a Banach space: X has (A) if $\text{NA}(X, Z)$ is dense in $\mathcal{L}(X, Z)$ for all Z ; and Y has (B) if $\text{NA}(W, Y)$ is dense in $\mathcal{L}(W, Y)$ for all W . Among many other results, he showed that reflexive spaces have (A) as does ℓ_1 , and that c_0 , ℓ_∞ and finite dimensional polyhedral spaces are examples of Banach spaces with property (B) and, finally, that there are Banach spaces (X, Y) such that $\text{NA}(X, Y)$ is not dense in $\mathcal{L}(X, Y)$. Major progress was made by Bourgain [9] who proved that spaces with the RNP have property (A) and provided a certain converse result. The problem of whether Hilbert spaces have property (B) was left open in [24], and it was solved only 25 years later by Gowers [17, Appendix], who showed that none of the spaces ℓ_p for $1 < p < \infty$ have (B). This result was pushed out

Figure 3: arXiv:1905.08272, page 1: finite-dimensional polyhedral spaces are listed among Lindenstrauss's examples with property B.

Proof intuition for the known subcase

If Y is finite-dimensional and polyhedral, then the dual ball B_{Y^*} has finitely many extreme points and

$$\|y\| = \max_{\phi \in \text{ext} B_{Y^*}} \phi(y).$$

Hence the norm of an operator $T: X \rightarrow Y$ is the maximum of finitely many scalar operator norms $\|\phi T\|$. Choose an extreme functional ϕ_0 at which this maximum is attained. Since ϕ_0 is a vertex of a polytope, one can expose it by a vector $y_0 \in S_Y$, so all other non-opposite extreme functionals have $|\phi(y_0)| < 1$.

The proof then makes the ϕ_0 -coordinate slightly dominant and uses the Bishop-Phelps theorem only in that scalar coordinate. The exposing gap prevents all other extreme coordinates from becoming dominant after the small rank-one correction, so the resulting vector-valued operator attains its full norm wherever the corrected scalar functional attains its norm.

Theorem 1. *Let Y be a real finite-dimensional polyhedral Banach space. Then for every real Banach space X , $\text{NA}(X, Y)$ is dense in $\mathcal{L}(X, Y)$.*

Proof. The case $Y = \{0\}$ is trivial, so assume $Y \neq \{0\}$. Let

$$F = \text{ext} B_{Y^*}.$$

Since Y is finite-dimensional and polyhedral, F is finite and symmetric, and $\|y\| = \max_{\phi \in F} \phi(y)$ for all $y \in Y$.

Fix $T \in \mathcal{L}(X, Y)$ and $\varepsilon > 0$. If $T = 0$, then T itself attains its norm. Assume $T \neq 0$, and put $M = \|T\|$. Choose $\phi_0 \in F$ such that

$$\|\phi_0 T\| = M.$$

operators [21] (including $X = C_0(L)$ and $X = L_1(\mu)$), it was proved in [29] that there exists a compact operator between certain Banach spaces which cannot be approximated by norm attaining operators. For more information and background on the topic of norm attaining operators we refer to the survey papers [3] and [30].

One should observe that none of the negative results summed up above says anything about operators of finite rank. Actually, it is one of the major open questions in the theory whether all finite dimensional Banach spaces have Lindenstrauss's property (B); equivalently, whether every finite rank operator between Banach spaces can be approximated by (finite rank) norm attaining operators. The aim of our paper is to contribute to this problem, in particular for rank-two operators.

In the case of linear functionals, it is clear from the Hahn-Banach theorem that $\text{NA}(X)$, the set of norm attaining functionals, is always non-empty. For operators

Figure 4: arXiv:1905.08272, page 2: the full finite-dimensional property B problem remains open.

Because ϕ_0 is a vertex of the polytope B_{Y^*} , there exists $y_0 \in Y$ exposing ϕ_0 . After scaling, we may assume

$$\phi_0(y_0) = 1 \quad \text{and} \quad \phi(y_0) < 1 \quad (\phi \in F, \phi \neq \phi_0).$$

Then $y_0 \in S_Y$, and

$$c := \max\{|\phi(y_0)| : \phi \in F, \phi \neq \pm\phi_0\} < 1.$$

Choose $\eta > 0$ and $\delta > 0$ so small that

$$\eta M + \delta < \varepsilon \quad \text{and} \quad (1 - c)\eta M > (1 + c)\delta.$$

By the scalar Bishop–Phelps theorem, there is a norm-attaining functional $g \in X^*$ such that

$$\|g - (1 + \eta)\phi_0 T\| < \delta.$$

Define $S : X \rightarrow Y$ by

$$Sx = Tx + (g(x) - \phi_0 T(x))y_0.$$

Then

$$\|S - T\| \leq \|g - \phi_0 T\| \leq \eta M + \delta < \varepsilon.$$

Moreover, $\phi_0 S = g$ and $(-\phi_0)S = -g$. For $\phi \in F \setminus \{\pm\phi_0\}$,

$$\|\phi S\| \leq \|\phi T\| + |\phi(y_0)| \|g - \phi_0 T\| \leq M + c(\eta M + \delta).$$

The choice of δ gives

$$M + c(\eta M + \delta) < (1 + \eta)M - \delta \leq \|g\|.$$

Thus

$$\|S\| = \max_{\phi \in F} \|\phi S\| = \|g\|.$$

Since g attains its norm, choose $x_0 \in S_X$ with $|g(x_0)| = \|g\|$. If $g(x_0) = \|g\|$, then

$$\|Sx_0\| \geq \phi_0(Sx_0) = \|g\| = \|S\|.$$

If $g(x_0) = -\|g\|$, replace x_0 by $-x_0$. Hence $S \in \text{NA}(X, Y)$. Since T and ε were arbitrary, $\text{NA}(X, Y)$ is dense in $\mathcal{L}(X, Y)$. $\square$

Verification notes and limitations

The proof is purely analytic and uses only the scalar Bishop–Phelps theorem and the finite exposed-vertex structure of the dual unit ball of a real polyhedral finite-dimensional space.

This is a literature-known proper subcase of Question 9. It does not cover real finite-dimensional spaces with smooth boundary, and in particular does not cover the two-dimensional real Hilbert space singled out by Martin. It also does not treat complex finite-dimensional spaces; the finite-vertex argument is intrinsically real-polyhedral.

No novelty claim is made. The supporting literature explicitly records the polyhedral subcase as known and explicitly says that the full finite-dimensional problem remains open.